



UNIVERSITÀ DEGLI STUDI DI MILANO

FACOLTÀ DI SCIENZE POLITICHE, ECONOMICHE E SOCIALI

FACOLTÀ DI SCIENZE E TECNOLOGIE

Master's Degree in Data Science for Economics

The computational narrative fingerprint: a structural NLP analysis of conflict narratives in online news media

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Research questions

Existing methods measure **what** conflict narratives say. This thesis measures **how** they are structured.

Is **agency** distributed asymmetrically across conflict sides, and in what structural form?

Do opposing outlets inhabit divergent **semantic spaces** across shared conflict vocabulary?

Is the **in-group** moral advantage symmetric across opposing outlets within each theater?

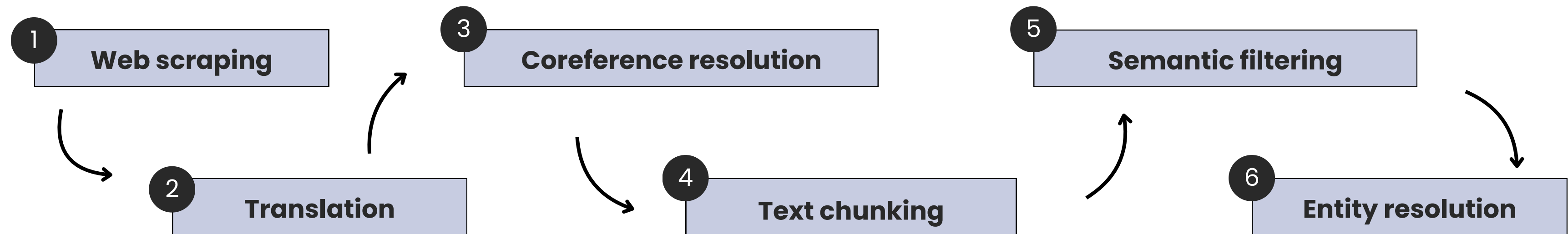
Do these structural patterns generalize across two **geopolitically distinct** conflict theaters?

Corpus overview

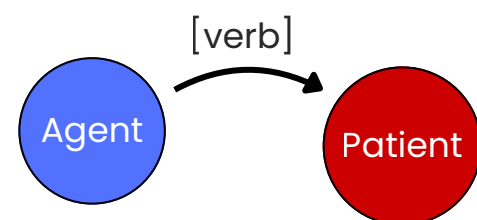
OUTLET	SIDE	ARTICLES	TOTAL TEXT CHUNKS
<i>Komsomolskaya Pravda</i>		313	1269
<i>Russia Today</i>		150	
<i>Ukrainska Pravda</i>		146	1181
<i>Liga.net</i>		80	
<i>Jerusalem Post</i>		236	1766
<i>Ynet</i>		236	
<i>Ynet Global</i>		102	
<i>Al-Quds</i>		340	1924

The conflicts in Palestine and in Ukraine are two of the most extensively documented and polarized conflicts of our time. This ensured sufficient **data volume** and **narrative depth**.

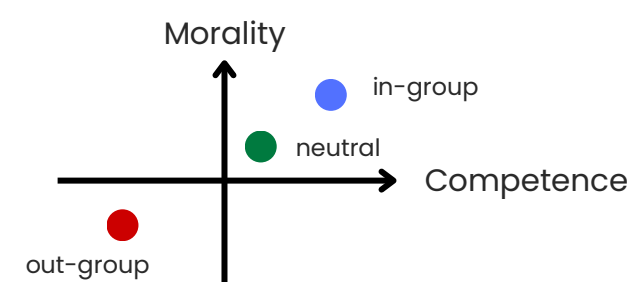
Analysis pipeline



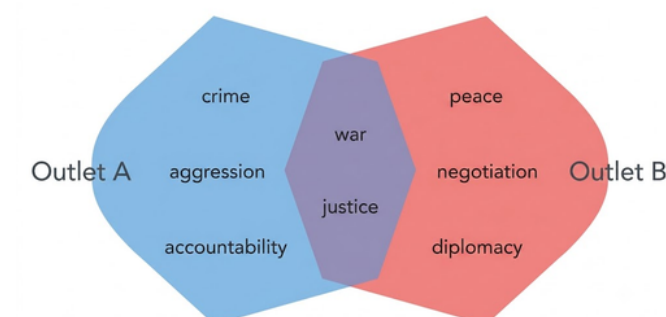
Agency and narrative networks



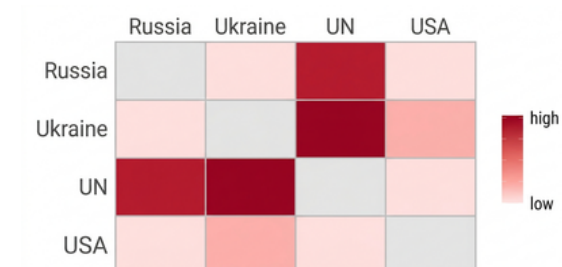
Character framing



Semantic divergence



Co-occurrence analysis



Agency extraction and narrative networks

The **Agency Ratio** is the proportion of an entity's narrative appearances in which it occupies the agent role rather than the patient role, computed as the count of agent appearances over the sum of agent and patient appearances.

The **Net Sentiment Agency** is the outgoing sentiment mass minus the incoming mass, normalized by total weight. *mean sentiment* weights each edge w .

$$AR(n) = \frac{AGENT(n)}{AGENT(n) + PATIENT(n)}$$

$$NSA(n) = \frac{\sum_{v \in \text{succ}(n)} \bar{s}_{nv} \cdot w_{nv} - \sum_{u \in \text{pred}(n)} \bar{s}_{un} \cdot w_{un}}{\sum_{v \in \text{succ}(n)} w_{nv} + \sum_{u \in \text{pred}(n)} w_{un}}$$

Each extracted triplet (Agent, Verb, Patient) becomes a directed, **sentiment-weighted edge** $n \rightarrow v$, with weight proportional to action count.

Hyperlink-Induced Topic Search (**HITS**) assigns two complementary scores to each node.

A node's hub score grows when it directs actions toward high-authority targets; its authority score grows when it is targeted by high-hub actors.

High $h \rightarrow$ narrative driver

High $a \rightarrow$ narrative patient

Character framing, semantic divergence and co-occurrence

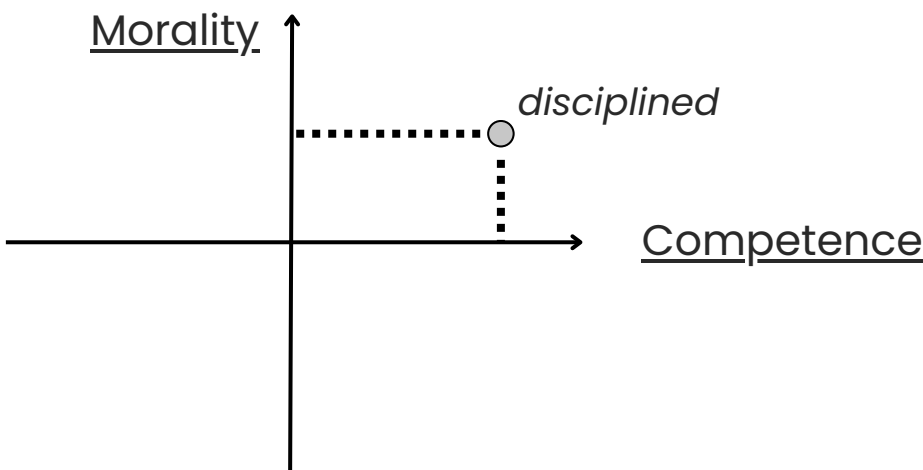
Two semantic axes are built from spaCy's pre-trained GloVe vectors. Each axis is the **vector difference** between the mean of a high-pole seed set and a low-pole seed set:

$$\vec{a}_{\text{competence}} = \mu(\mathbf{v}_{\text{powerful}}, \mathbf{v}_{\text{capable}}, \dots) - \mu(\mathbf{v}_{\text{weak}}, \mathbf{v}_{\text{incompetent}}, \dots)$$

$$\vec{a}_{\text{morality}} = \mu(\mathbf{v}_{\text{righteous}}, \mathbf{v}_{\text{just}}, \dots) - \mu(\mathbf{v}_{\text{cruel}}, \mathbf{v}_{\text{corrupt}}, \dots)$$

Each *adjective* syntactically linked to a target entity is projected onto both axes as a *scalar projection*:

$$\text{score}(\text{adj}, \vec{a}) = \frac{\vec{v}_{\text{adj}} \cdot \vec{a}}{\|\vec{a}\|}$$



Lexical divergence

z-scored frequency ratio; identifies statistically distinctive words per outlet; a Dirichlet prior prevents log-zero on unseen words.

Contextual divergence

Jaccard distance between co-occurrence neighbor sets in shared GloVe space; captures how differently outlets frame the same concept.

Entity pairs **co-occurring** in the same sentence are quantified via *two complementary measures*:

$$\text{Jaccard}(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad \text{PMI}(A, B) = \log \frac{P(A, B)}{P(A) \cdot P(B)}$$

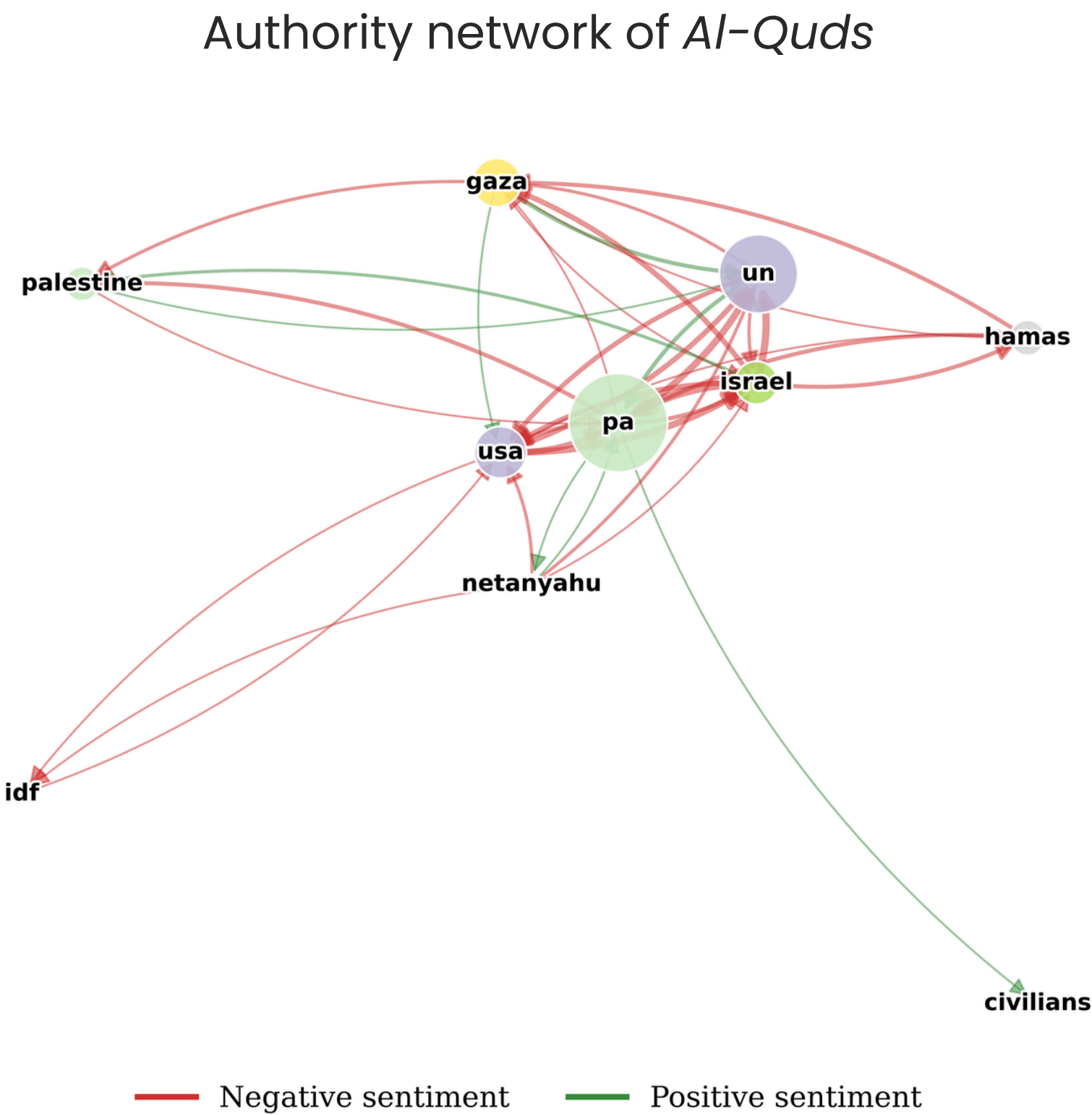
Agency across both theaters

In the Russia-Ukraine theater, Russian outlets assign Russia a positive NSA while Ukrainian outlets assign it a negative one. Sentiment valence toward Russia **inverts bilaterally** by outlet nationality.

In the Israel-Palestine theater, Israel is the **top narrative hub** in every outlet, including Al-Quds, and its actions carry negative or near-neutral valence even in Israeli outlets.

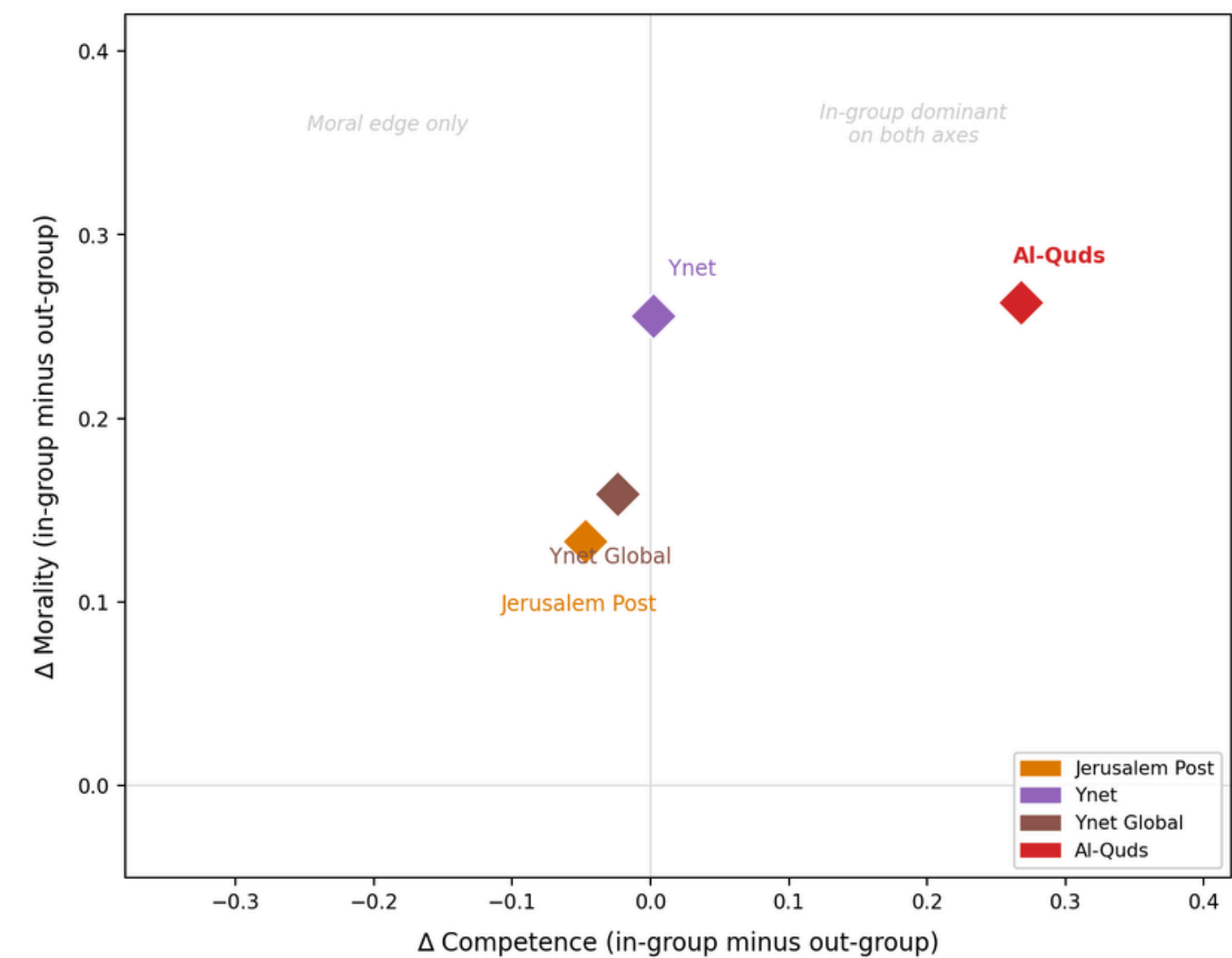
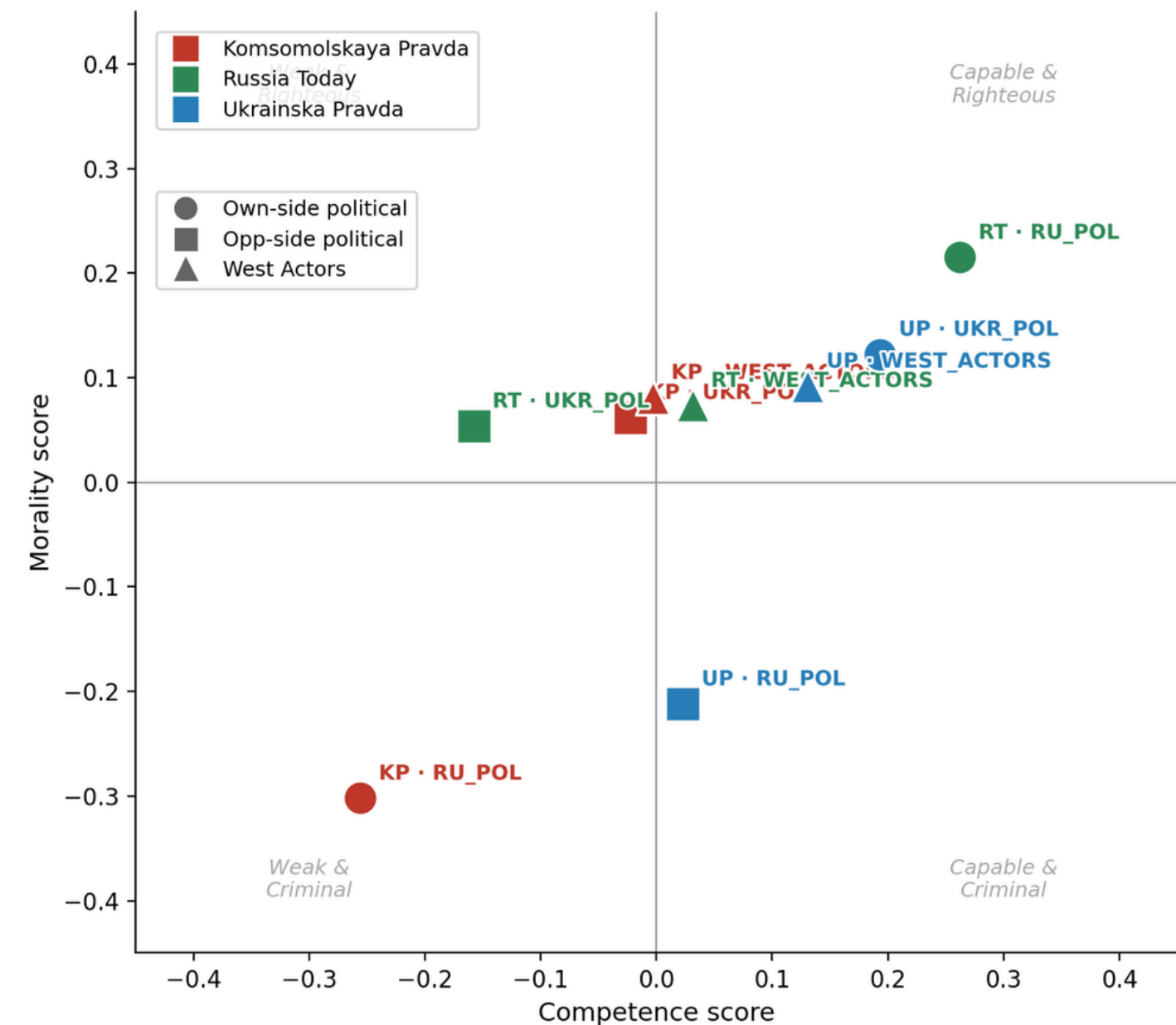
OUTLET	RUSSIA NSA
<i>Kom. Pravda</i>	+0.052
<i>Russia Today</i>	+0.262
<i>Ukr. Pravda</i>	-0.236
<i>Liga.net</i>	-0.448

OUTLET	TOP HUB
<i>Al-Quds</i>	Israel
<i>Jerusalem Post</i>	Israel
<i>Ynet</i>	Israel
<i>Ynet Global</i>	Israel



Character framing across both theaters

Russia Today frames Russian leadership as capable and righteous, Ukrainian leadership as weak: the **clearest mirror image** in the theater. *Ukrainska Pravda* contests Russia on the morality axis, framing Russia as criminal, not incompetent.



All four outlets assign **higher morality to their in-group**. The disagreement is not whether one side is more righteous, but by how much. *Al-Quds* is the only outlet with both differentials strongly positive.

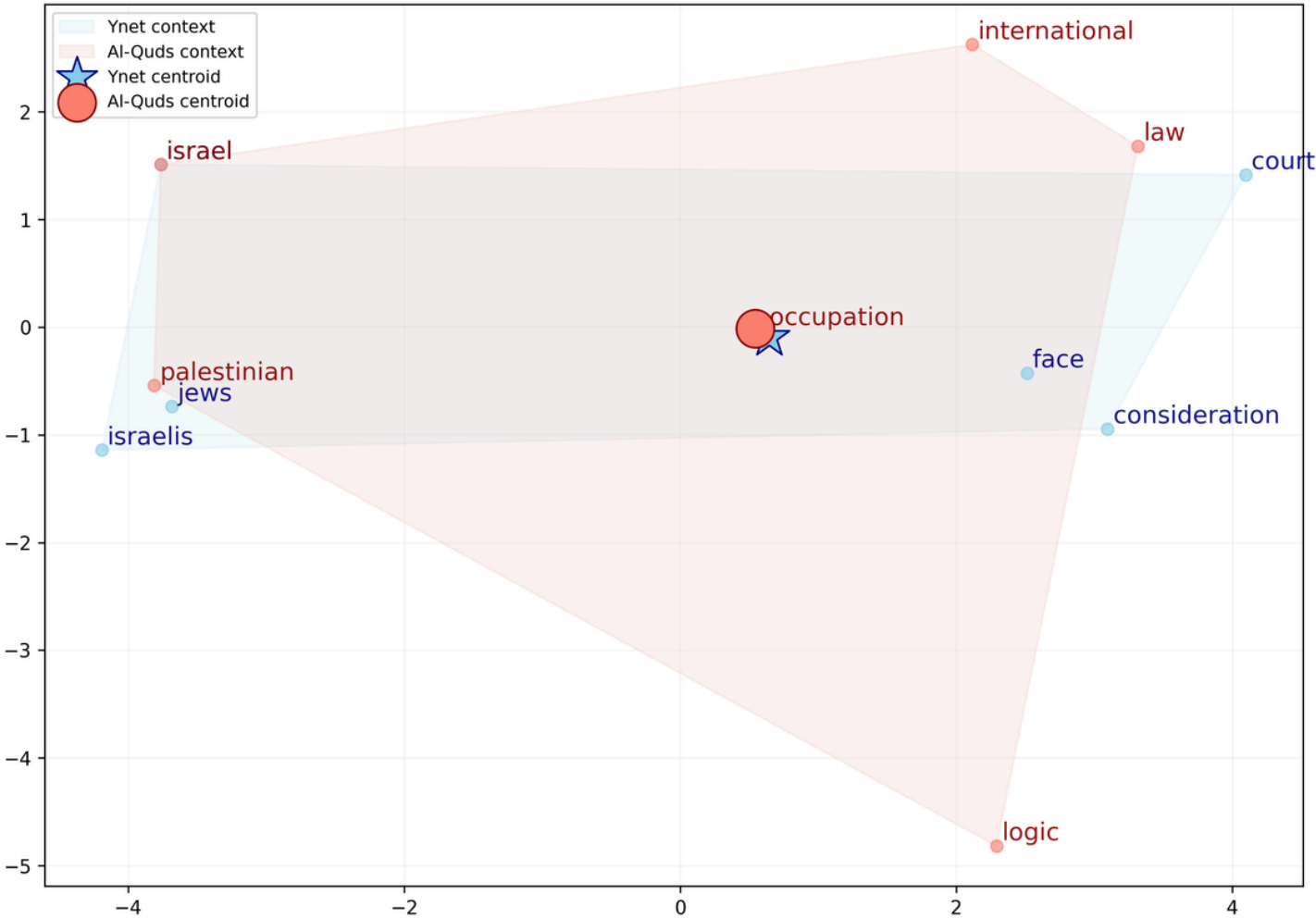
Semantic divergence across both theaters

Komsomolskaya Pravda narrates the conflict as a sequence of diplomatic events around named actors. *Ukrainska Pravda* narrates it as a legal crisis.

Kom. Pravda	Ukrainska Pravda
Trump	War
Zelensky	Crime
Meeting	Law
Talk	Aggression

Most distinctive vocabulary per outlet (log-odds, ordered by z-score)

In both theaters, **"Justice"** records the highest semantic divergence among 9 tracked concepts: opponents share the word but not its neighboring vocabulary.



Ynet vs Al-Quds semantic map for the word *"Justice"*: high Jaccard distance.

"Occupation" appears as an Al-Quds exclusive neighbor across four tracked concepts, and never in *Ynet*.

Four questions, four answers

RQ 1

Agency is **asymmetric** in every outlet. Form differs: *sentiment-valence inversion* in Russia-Ukraine, *hub-identity invariance* in Israel-Palestine.

RQ 3

Opposing outlets construct **divergent** semantic worlds around shared vocabulary. “Justice” records maximum divergence in both theaters.

RQ 2

In-group moral advantage holds in 6 of 8 outlets. Consistent in Israel-Palestine; partial in Russia-Ukraine (2/4).

RQ 4

Three patterns **recur** across both theaters and both sides: agency asymmetry, in-group moral advantage in 6 of 8 outlets, and normative concepts like “justice” diverging in meaning.

Limitations, contributions and future directions

Limitations

Corpus size is modest overall, which may limit the statistical power of some analyses.

Articles translation might attenuate rhetorical features.

Main contributions

CNF architecture generalizes by design: no pipeline redesign needed for new conflicts

A fully interpretable alternative to LLM-based approaches: every score traces back to a named linguistic feature, not a black box.

Refines *Social Identity Theory*: structural role and moral valence can move independently, not just together.

Future directions

Extend to a third conflict theater to test whether the patterns generalize beyond two cases.

Develop the proof-of-concept classifier into a validated tool, and benchmark it against LLMs to see what structural interpretability adds.

thank you.